FLEET TEAM DIGEST

737MAX-FTD-32-17011

Issue Title : Main Landing Gear Shimmy

Airplane Model	ATA	Minor Model(s)
737MAX	3210-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737	9E991

Originated Date	Last Revised Date	Created On
05/02/2017	05/02/2017	04/21/2017

Estimated Completion Date	Next Update Date

Applicability

All 737MAX airplanes.

Description

The 737NG fleet has sporadically experienced a small number of main landing gear shimmy events since shimmy dampers were installed. Shimmy events are characterized by extreme vibration from one (or both) of the main landing gears. The vibration starts at touchdown and usually continues until the airplane is fully or near fully stopped. Shimmy events usually result in damage to the affected landing gear, dislodged ceiling panels, and dropped passenger service units (PSU's). In a few of the more severe shimmy events, other damage has been reported to the wings and to the aft pressure bulkhead.

This article discusses shimmy, the causes of shimmy, and recommended actions operators can take to reduce the likelihood of shimmy.

Background

Shimmy is a torsional vibration mode of the landing gear where the inner cylinder rotates (oscillates) relative to the outer cylinder (see attached graphic). To prevent this vibration mode, most large airplanes with single-axle main landing gears have a hydraulic damper to prevent shimmy. The damper is connected between the upper and lower torsion links and allows a small, but highly damped, motion to occur around the torsional axis of the gear.

Since incorporation of the dampers, the rate of shimmy has dropped dramatically and now shimmy events are extremely rare. However, shimmy events still occur occasionally for a variety of reasons.

The vibration frequency of shimmy on a 737 is about 15 Hz. In order for shimmy to occur, the landing gear must have a force applied to it that excites this torsional vibration mode. It is theorized that the force needed to initiate shimmy is an alternating force, such as if one tire touches down causing a twisting motion of the inner cylinder in one direction and then a fraction of a second later, the second tire touches down, causing the inner cylinder to twist in the opposite direction. If the timing between the first tire and second tire contacting the runway is similar to the shimmy frequency, the gear may start to oscillate in the shimmy mode.

Due to the geometry of the torsion links, the shimmy damper is most effective when the landing gear strut is compressed in the ground mode. With the shock strut fully or near fully extended, the torsion links hang in a near vertical position which gives the damper less mechanical advantage to perform its function. Low sink rate landings leave the torsion links in this position longer than normal sink rate landings. As such, landing technique can affect the likelihood of a shimmy event.

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Main gear shimmy on a takeoff roll has never been reported on the 737 models, most likely since severe twisting forces are never applied to the gear during a takeoff and the torsion links are in a more advantageous orientation (ground mode).

Status

Closed.

Interim Action

N/A

Final Action

Boeing has been studying each shimmy event in an attempt to understand the root cause(s) and to incorporate any corrective actions or recommendations.

Operator Action

If an operator suspects that a shimmy event has occurred, it is important to perform the conditional inspections contained in AMM 05-51-68.

While the 737 shimmy dampers have been very successful at preventing shimmy, there are problems that can arise that render the dampers ineffective. Detailed studies of the 737NG shimmy events have revealed several root causes. There are also some theoretical root causes which can increase the likelihood of a shimmy event but have not been specifically identified as causal in any of the 737NG shimmy events.

FINDINGS FROM IN-SERVICE EVENTS

1) Installation error. There are two cylindrical sleeves that are installed on the torsion link apex pin during damper installation (a 'straight' sleeve and the 'tapered' sleeve). It has been found that these sleeves have been inadvertently reversed during damper installation on a few of the shimmy events. Reversing the orientation of these sleeves results in excessive torsional freeplay of the landing gear inner cylinder, which can result in a shimmy event. Refer to AMM 32-11-81, Figure 401 to see the correct sleeve orientation.

2) Landings with extremely low sink rates are more likely to experience shimmy than firmer landings since the torsion links remain in a position where the damper has less mechanical advantage for longer periods of time. We suggest following the landing procedure in the Flight Crew Training Manual (FCTM) and ensure prompt speedbrake deployment.

3) In a few events, the compensator portion of the shimmy damper separated from the damper piston housing, thereby preventing proper damper operation. It is believed that the compensators became detached from the damper due to contact with a foreign object on a previous flight. Refer to 737NG-FTD-32-10001 for details.

4) Incorrectly torqued torsion link apex nut. In one event, it was found that the nut at the torsion link apex was incorrectly torqued. This resulted in excessive torsional freeplay in the landing gear. The correct torquing procedure is shown in AMM 32-11-81/401.

OTHER POSSIBLE CAUSES OF SHIMMY

a) Wear or freeplay in the torsion link bushings (where the torsion links connect to the outer and inner cylinder) has the potential to result in excessive torsional freeplay. Wear at these locations allows un-damped torsional freeplay to exist in the landing gear which will greatly increase the likelihood of shimmy.

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b) Air in the damper. If the damper is not properly bled of air, the shimmy damper will not function correctly.

c) An over-serviced shock strut results in a torsion link orientation where the shimmy damper has less mechanical advantage to prevent shimmy.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737NG-FTD-32-10001		

Related Categories

- In Service Occurrences : null

Attachments:

a) Shimmy mode vibration

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